CORE PURE / COMPLEX NUMBERS

# Complex Roots and Geometry Problems

Mark scheme

13 questions · 127 marks

## Questions


| Question | Marks by part | Marks | Page |
|---|---|---|---|
| Question 1 | (a)(i) 2 (a)(ii) 2 (b) 4 | 8 | 4 |
| Question 2 | (a) 5 (b) 2 (c) 4 | 11 | 6 |
| Question 3 | (a) 4 (b) 5 | 9 | 8 |
| Question 4 | (a) 2 (b) 2 (c) 2 (d) 2 (e) 2 | 10 | 9 |
| Question 5 | | 5 | 11 |
| Question 6 | (a) 6 (b) 4 | 10 | 12 |
| Question 7 | (a) 3 (b) 4 | 7 | 13 |
| Question 8 | (a) 2 (b) 1 (c) 2 (d) 3 (e) 4 | 12 | 14 |
| Question 9 | (a) 2 (b) 2 (c)(i) 1 (c)(ii) 4 | 9 | 16 |
| Question 10 | (i) 5 (ii) 2 (iii) 3 (iv) 3 | 13 | 17 |
| Question 11 | (a)(i) 1 (a)(ii) 3 (a)(iii) 3 (b)(i) 2 (b)(ii) 2 (b)(iii) 3 | 14 | 19 |
| Question 12 | (a) 2 (b) 4 (c) 2 | 8 | 21 |
| Question 13 | (i) 2 (ii) 6 (iii) 1 (iv) 2 | 11 | 22 |
| | **Total** | **127** | |

# Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

### Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

| Code | Meaning |
|---|---|
| **M1** | Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted. |
| **dM1** | Dependent method: as M1, but only if the method mark before it was earned, unless its note names another. |
| **A1** | Accuracy: a correct answer or step. It needs the method mark it follows. |
| **A1ft** | As A1, following through from an earlier answer that the note names. |
| **A1 AG** | The last mark of a part whose answer is printed in the question: complete working, with no errors. |
| **B1** | A correct value, fact or statement. It needs no method mark. |
| **B1ft** | As B1, following through from an earlier answer that the note names. |
| **E1** | A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark. |

### Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes.

| | | | |
|---|---|---|---|
| **ft** | follow through | **AG** | answer given in the question |
| **cao** | correct answer only, with no follow-through | **cso** | correct solution only: no errors in the part |
| **oe** | or an equivalent form | **awrt** | any answer that rounds to this value |
| **soi** | seen, or implied by later working | **isw** | ignore working after a correct answer |
| **SC** | special case: credit outside the main method | **their** | the student's own earlier value |
| **condone** | a slip that doesn't cost the mark | [ ] | working the mark doesn't need |
| **s.f.** | significant figures | **d.p.** | decimal places |

### Reading the scheme

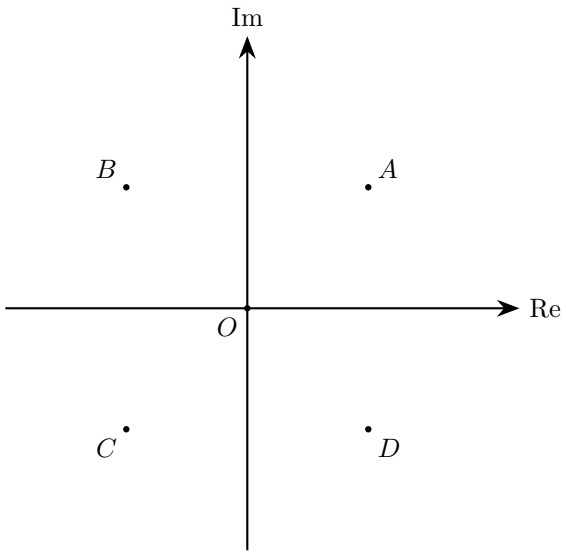
- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- **[*n*]** ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

### Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

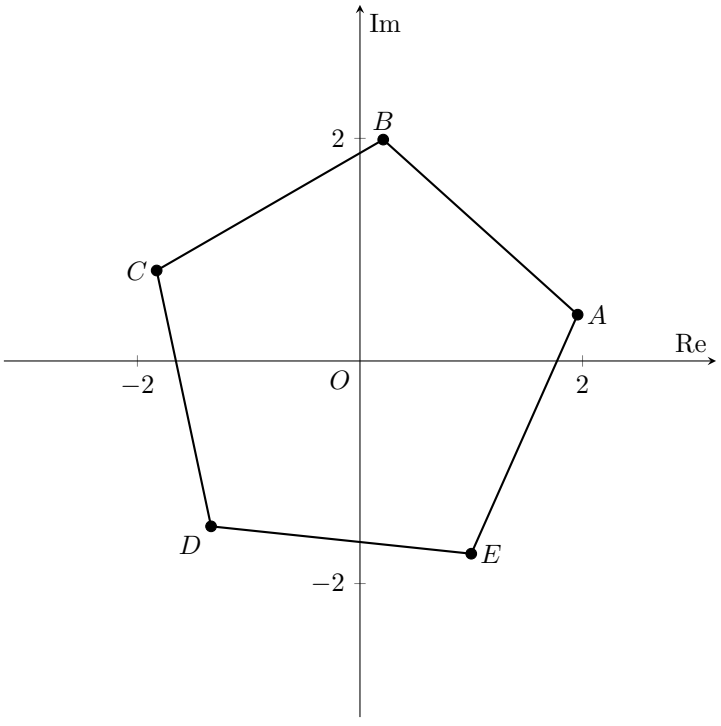
### Value of $g$

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state $g$, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **1(a)(i)** |  | | |
| | *Labelled point* | **B1** | $B$ and $D$ at the quarter-turn rotations of $A$ about $O$, $B$ anticlockwise from $A$. Condone small inaccuracy in position. |
| | *Symmetry* | **B1** | $C$ opposite $A$ through $O$, labelled $C$. |
| | | **[2]** | |
| **1(a)(ii)** | $z_1 + z_1 e^{\frac{1}{2}\pi i} + z_1 e^{\pi i} + z_1 e^{\frac{3}{2}\pi i}$<br>$= z_1 + iz_1 - z_1 - iz_1$ | **M1** | Evaluates the three exponentials, directly or by $\cos\theta + i\sin\theta$, at least two correctly, and sums the resulting multiples of $z_1$. |
| | $= z_1(1 + i - 1 - i) = 0$ | **A1 AG** | All three values correct and the terms shown to cancel to 0 without error. |
| | *Alternative:* Geometric series | | |
| | $z_1\left(1 + e^{\frac{1}{2}\pi i} + e^{\pi i} + e^{\frac{3}{2}\pi i}\right)$<br>$= z_1 \dfrac{1 - \left(e^{\frac{1}{2}\pi i}\right)^4}{1 - e^{\frac{1}{2}\pi i}}$ | **M1** | Uses the sum of four terms of a geometric series with ratio $e^{\frac{1}{2}\pi i}$. Condone the factor $z_1$ omitted here. |
| | $\left(e^{\frac{1}{2}\pi i}\right)^4 = e^{2\pi i} = 1 \Rightarrow$ sum $= 0$ | **A1 AG** | Shows the ratio to the fourth power is 1 and concludes the sum is 0, with the denominator nonzero. |
| | | **[2]** | |
| **1(b)** | $-16 = 16e^{\pi i}$<br>$= 16e^{(\pi + 2k\pi)i}$ | **B1** | Accept modulus–argument form, and the four arguments listed in place of $\pi + 2k\pi$. |

*Question 1 continued*

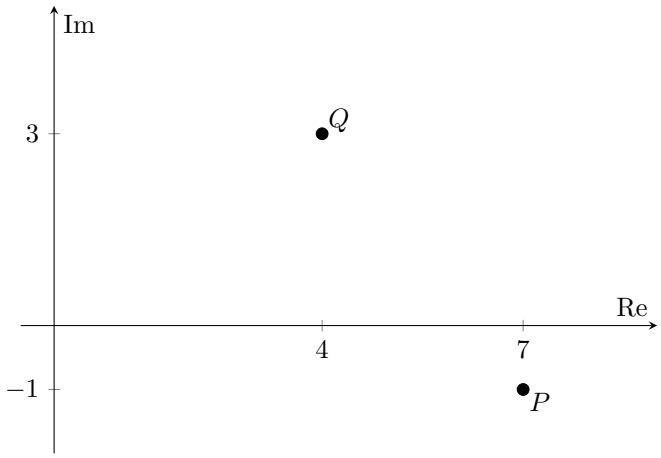
| Question | Scheme | Marks | Notes |
|---|---|---|---|
| | $z = 2e^{\frac{\pi+2k\pi}{4}i}$, $k = 0, 1, 2, 3$ | **M1** | Takes the fourth root of their modulus and divides their arguments by 4, giving at least two distinct roots in any form. |
| | $z = \sqrt{2} + \sqrt{2}\,i,\ -\sqrt{2} + \sqrt{2}\,i$ | **A1** | Any two correct roots, in any form. |
| | $z = \pm\sqrt{2} \pm \sqrt{2}\,i$ (all four) | **A1** | All four in the form $a + bi$ with exact $a$ and $b$, and no others. Roots left in exponential form score A0. |
| | | **[4]** | |
| **Total for Question 1** | | **8** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **2(a)** | $\lvert w\rvert = \sqrt{16^2 + (16\sqrt{3})^2}$<br>$\arg w = \arctan \frac{16\sqrt{3}}{16}$ | **M1** | Attempts both the modulus and the argument of $16 + 16\sqrt{3}\,i$. Condone arithmetic slips. |
| | $16 + 16\sqrt{3}\,i = 32e^{\frac{\pi}{3}i}$ | **A1** | Or modulus 32 and argument $\frac{\pi}{3}$ (awrt 1.05) stated. |
| | $z = \text{"}32\text{"}^{\frac{1}{5}} e^{\frac{\text{"}\pi/3\text{"}+2k\pi}{5}i}$ | **M1** | Takes the fifth root of their modulus and uses arguments (their argument $+2k\pi$)/5, for general $k$ or at least two values, or rotates one root by $\frac{2\pi}{5}$. |
| | $2e^{\frac{\pi}{15}i},\ 2e^{\frac{7\pi}{15}i}, \ldots$ | **A1** | Any two correct roots, in any form. |
| | $2e^{\frac{\pi}{15}i},\ 2e^{\frac{7\pi}{15}i},\ 2e^{\frac{13\pi}{15}i},$<br>$2e^{\frac{19\pi}{15}i},\ 2e^{\frac{5\pi}{3}i}$ | **A1** | All five as $2e^{i\theta}$ with $0 \le \theta < 2\pi$, exact or awrt 0.209, 1.47, 2.72, 3.98, 5.24, and no others. A0 for any argument outside the range. |
| | | **[5]** | |
| **2(b)** |  | | |
| | *Shape* | **B1** | Pentagon centred at $O$ with vertices equally spaced on a circle of radius 2, the radius shown by a scale or label. |

*Question 2 continued*

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| | *Labelled point* | **B1** | Each vertex labelled with its root, or a letter identified with it, in its correct position, one near the positive real axis in the first quadrant. |
| | | **[2]** | |
| **2(c)** | Five congruent triangles with sides $2, 2$ and angle $\frac{2\pi}{5}$ at $O$ | **B1** | Angle at $O$ may be implied by its use in an area. |
| | $\frac{1}{2} \times 2 \times 2 \times \sin \frac{2\pi}{5}$ | **M1** | Uses $\frac{1}{2}ab\sin C$ with two sides equal to their modulus and their angle at $O$ as the included angle. |
| | $= 2\sin \frac{2\pi}{5}$ | **A1** | |
| | $5 \times 2\sin \frac{2\pi}{5} = 10\sin \frac{2\pi}{5} \Rightarrow k = 10$ | **A1** | cso. Area $10\sin \frac{2\pi}{5}$ or $k = 10$ stated; a decimal area alone scores A0. |
| | | **[4]** | |
| | **Total for Question 2** | **11** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **3(a)** | $(h-1)^2+4^2=(h-7)^2+2^2$<br>$\Rightarrow h=\ldots$ | **M1** | Equates the distances, or squared distances, from $(h,0)$ to $A$ and $B$, or intersects the perpendicular bisector of $AB$ with the real axis. Condone sign slips. |
| | $h=3$ | **A1** | |
| | $r^2=(\text{"}3\text{"}-1)^2+4^2$ $[=20]$ | **M1** | Uses Pythagoras for the distance from their real centre to $A$ or $B$. |
| | $\lvert z-3\rvert=2\sqrt{5}$ | **A1** | Accept $\sqrt{20}$ or awrt 4.47. Only the form $\lvert z-a\rvert=b$ scores; $(x-3)^2+y^2=20$ alone scores A0. |
| | | **[4]** | |
| **3(b)** | $(x-3)^2+y^2=20$<br>$(x-1)^2+(y-1)^2=25$ | **M1 A1** | **M1** Writes at least one circle as an equation in $x$ and $y$, using $z=x+iy$ and their $C$ of the form $\lvert z-a\rvert=b$. Condone sign slips.<br>**A1** Both correct, in any form. |
| | $4x-2y-12=0\Rightarrow y=2x-6$ | **M1** | Subtracts their equations to eliminate $x^2$ and $y^2$, reaching a linear equation in $x$ and $y$. Condone sign slips. |
| | $(x-3)^2+(\text{"}2x-6\text{"})^2=20$<br>$\Rightarrow x=\ldots$ | **dM1** | Substitutes their linear equation into either circle and solves the resulting quadratic in one variable by any method. |
| | $z_1=1-4i,\ z_2=5+4i$ | **A1** | Both, in either order, as complex numbers; coordinates alone score A0. |
| | | **[5]** | |
| | **Total for Question 3** | **9** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **4(a)** | $z_2 = \frac{(7-i)(1+i)}{(1-i)(1+i)} = \ldots$ | **M1** | Multiplies numerator and denominator by $1+i$ and expands, or equates real and imaginary parts of $(a+bi)(1-i) = 7-i$. Condone one slip. |
| | $z_2 = 4+3i$ | **A1** | |
| | | **[2]** | |
| **4(b)** |  | | |
| | *Labelled point* | **B1** | $P$ plotted and labelled at $(7,-1)$, in the fourth quadrant, with its position shown by scale or coordinates. |
| | *Labelled point* | **B1ft** | ft their $z_2$ from (a). $Q$ plotted and labelled at their point, positioned correctly relative to $P$. |
| | | **[2]** | |
| **4(c)** | $\overrightarrow{QO} \cdot \overrightarrow{QP} = (-4)(3) + (-3)(-4)$ | **M1** | Applies a perpendicularity test at $Q$ to $OQ$ and $QP$, such as a scalar product, a product of gradients or Pythagoras' converse. Condone sign slips. |
| | $= 0 \Rightarrow \angle OQP = \frac{\pi}{2}$ | **A1 AG** | Correct values giving 0, a gradient product of $-1$, or $OQ^2 + QP^2 = OP^2$, with a conclusion that the angle is a right angle. |
| | | **[2]** | |
| **4(d)** | Uses $OP$ as a diameter: $C = \frac{1}{2}(7-i) = \ldots$ | **M1** | Takes $C$ as the midpoint of $OP$, or solves for the point equidistant from $O$, $P$ and $Q$. |

*Question 4 continued*

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| | $C = \frac{7}{2} - \frac{1}{2}i$ | **A1** | oe as a complex number, such as $3.5 - 0.5i$. Coordinates $\left(\frac{7}{2}, -\frac{1}{2}\right)$ alone do not score. |
| | | **[2]** | |
| **4(e)** | $r = \frac{1}{2}\lvert OP\rvert = \frac{1}{2}\sqrt{7^2 + (-1)^2}$ | **M1** | Finds half of $\lvert OP\rvert$, or the distance from their $C$ to $O$, $P$ or $Q$, by Pythagoras. Condone sign slips. |
| | $r = \frac{5\sqrt{2}}{2}$ | **A1** | Any exact form, such as $\frac{\sqrt{50}}{2}$ or $\sqrt{12.5}$. |
| | | **[2]** | |
| | **Total for Question 4** | **10** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **5** | $A$, $B$, $C$: $z^3 - 1 = 0$ | **B1** | |
| | $P$, $Q$, $R$: $(z-2)^3 = 1$ | **M1** | Replaces $z$ by $z-2$ in their cubic, or forms a cubic from the three roots by factors or root sums and products. Need not be expanded. |
| | $z^3 - 6z^2 + 12z - 9 = 0$ | **A1** | Correct cubic, oe such as $(z-3)(z^2-3z+3) = 0$. |
| | $(z^3-1)(z^3-6z^2+12z-9) = 0$<br>$\Rightarrow z^6 - 6z^5 + \ldots = 0$ | **M1** | Multiplies their two cubics and expands. Condone slips in collecting terms. |
| | $z^6 - 6z^5 + 12z^4 - 10z^3 + 6z^2 - 12z + 9 = 0$ | **A1** | Any nonzero integer multiple accepted. |
| | **Total for Question 5** | **5** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **6(a)** | $z_A - z_G = 4 + 4i$ | **B1** | |
| | $z_B - z_G = (4+4i)\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)$<br>$= \ldots$ | **M1** | Multiplies their $z_A - z_G$ by $e^{\pm 2\pi i/3}$ in any form and expands. Condone a sign slip in the cosine or sine value. |
| | $z_B - z_G = (-2 - 2\sqrt{3}) + (2\sqrt{3} - 2)i$ | **A1** | |
| | $z_B = (-1 - 2\sqrt{3}) + (2\sqrt{3} - 4)i$ | **A1** | Exact, in the form $a + bi$. $B$ and $C$ may be interchanged. |
| | $z_C = (1 - 2i) + (4 + 4i)\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)$<br>$= \ldots$ | **M1** | Rotates their $z_A - z_G$ through the other angle and adds $z_G$, or uses $z_C = 3z_G - z_A - z_B$ with their $z_B$. |
| | $z_C = (-1 + 2\sqrt{3}) - (4 + 2\sqrt{3})i$ | **A1** | Exact, in the form $a + bi$. |
| | | **[6]** | |
| **6(b)** | $z_D = \frac{1}{3}(z_G + z_B + z_C)$, $z_B + z_C = 3z_G - z_A$<br>$\Rightarrow z_D = \frac{1}{3}(4z_G - z_A)$ | **M1** | Writes $z_D$ as the mean of $z_G$, $z_B$, $z_C$, then simplifies using $z_A + z_B + z_C = 3z_G$ or evaluates with their (a) values. |
| | $z_D - z_G = -\frac{1}{3}(z_A - z_G)$, similarly for $E$, $F$<br>$\Rightarrow$ area factor $\left(\frac{1}{3}\right)^2 = \frac{1}{9}$ | **A1** | Lengths scaled by $\frac{1}{3}$ about $G$, sign not needed, with $E$ and $F$ by symmetry, and area factor $\frac{1}{9}$. |
| | $GA = \lvert 4 + 4i \rvert = 4\sqrt{2}$<br>Area $ABC = 3 \times \frac{1}{2}(4\sqrt{2})^2 \sin\frac{2\pi}{3}$<br>$= \ldots$ $[= 24\sqrt{3}]$ | **M1** | Uses three triangles $\frac{1}{2}r^2 \sin\frac{2\pi}{3}$ with their $r = \lvert z_A - z_G \rvert$, or $\frac{\sqrt{3}}{4}s^2$ with their side $s = \lvert z_A - z_B \rvert$. |
| | Area $DEF = \frac{1}{9} \times 24\sqrt{3} = \frac{8\sqrt{3}}{3}$ | **A1** | Exact, oe such as $\frac{8}{\sqrt{3}}$. |
| | | **[4]** | |
| | **Total for Question 6** | **10** | |

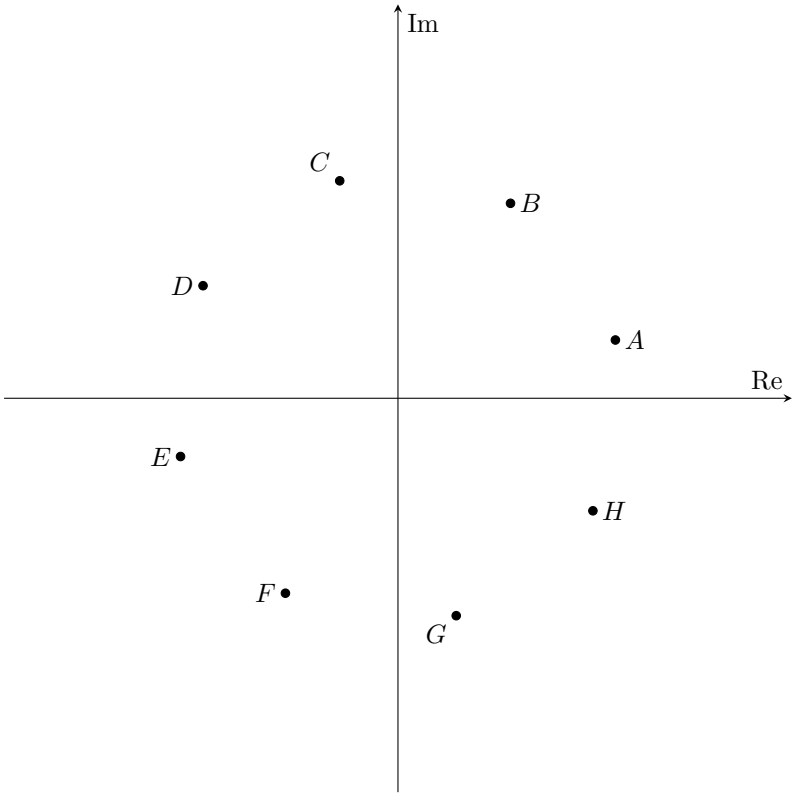
| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **7(a)** | $\lvert w-(1+i)\rvert^2 = \big(w-(1+i)\big)\big(w^*-(1-i)\big)$<br>$= \ldots$ | **M1** | Multiplies $w-(1+i)$ by its attempted conjugate and expands, or an equivalent Cartesian method. Condone sign slips. |
| | $= ww^* - (1-i)w - (1+i)w^* + (1+i)(1-i)$ | **A1** | Correct four-term expansion, oe. |
| | $ww^* = \lvert w\rvert^2 = 1$, $(1+i)(1-i) = 2$<br>$\Rightarrow \lvert w-(1+i)\rvert^2 = 3-(1-i)w-(1+i)w^*$ | **A1 AG** | Uses $ww^* = 1$ from $\lvert w\rvert = 1$ and $(1+i)(1-i) = 2$, reaching the printed result without error. |
| | | **[3]** | |
| **7(b)** | $A_{2r-1}$: $w_r = e^{\frac{2(r-1)\pi}{5}i}$, the fifth roots of unity<br>$P$: $1+i$ | **B1** | Needs both the odd-numbered vertices as the fifth roots of unity, or their five arguments, and $P$ as $1+i$. |
| | $\sum_{r=1}^{5}(PA_{2r-1})^2 = \sum_{r=1}^{5}\big(3-(1-i)w_r-(1+i)w_r^*\big)$<br>$= 15-(1-i)\sum w_r-(1+i)\sum w_r^*$ | **M1** | Sums the result of (a) over the five odd-numbered vertices, separating the sums of $w_r$ and $w_r^*$. Condone the constant left as $\sum 3$. |
| | $\sum w_r = 0$, $\sum w_r^* = 0$ | **B1** | Both sums stated or shown to be zero. |
| | $\sum_{r=1}^{5}(PA_{2r-1})^2 = 15$ | **A1** | cso. |
| | | **[4]** | |
| | **Total for Question 7** | **7** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **8(a)** | $\alpha^5 = \cos\left(5 \times \frac{3\pi}{5}\right) + i\sin\left(5 \times \frac{3\pi}{5}\right)$<br>$= \cos 3\pi + i\sin 3\pi$ | **M1** | Applies de Moivre's theorem to $\alpha^5$, reaching argument $3\pi$ or an equivalent. |
| | $\alpha^5 = -1$, so $\alpha$ is a root | **A1 AG** | Evaluates $\cos 3\pi = -1$ and $\sin 3\pi = 0$ and concludes that $\alpha$ is a root. |
| | | **[2]** | |
| **8(b)** | $\cos\frac{\pi}{5} + i\sin\frac{\pi}{5}$, $-1$,<br>$\cos\frac{7\pi}{5} + i\sin\frac{7\pi}{5}$, $\cos\frac{9\pi}{5} + i\sin\frac{9\pi}{5}$ | **B1** | All four, in any correct form, with arguments in any range such as $-\pi < \theta \leq \pi$. |
| | | **[1]** | |
| **8(c)** | $\alpha^5 + 1 = (\alpha+1)(\alpha^4 - \alpha^3 + \alpha^2 - \alpha + 1)$ | **M1** | Factorises $z^5 + 1$ with factor $z + 1$, or sums the geometric series with ratio $-\alpha$. Expanding the printed product to $\alpha^5 + 1$ also scores. |
| | $\alpha^5 + 1 = 0$ and $\alpha \neq -1$<br>$\Rightarrow \alpha^4 - \alpha^3 + \alpha^2 - \alpha + 1 = 0$ | **A1 AG** | Uses $\alpha^5 + 1 = 0$ and states $\alpha \neq -1$ before concluding. A0 if $\alpha \neq -1$ is not stated. |
| | | **[2]** | |
| **8(d)** | $\alpha^2 - \alpha + 1 - \frac{1}{\alpha} + \frac{1}{\alpha^2} = 0$ | **M1** | Divides the quartic by $\alpha^2$, or multiplies an expansion of the printed LHS by $\alpha^2$. Condone one slip. |
| | $\left(\alpha + \frac{1}{\alpha}\right)^2 = \alpha^2 + 2 + \frac{1}{\alpha^2}$<br>$\Rightarrow \left(\alpha + \frac{1}{\alpha}\right)^2 - 2 - \left(\alpha + \frac{1}{\alpha}\right) + 1 = 0$ | **M1** | Uses the expansion of $\left(\alpha + \frac{1}{\alpha}\right)^2$ to replace $\alpha^2 + \frac{1}{\alpha^2}$, or to expand the printed LHS. |
| | $\left(\alpha + \frac{1}{\alpha}\right)^2 - \left(\alpha + \frac{1}{\alpha}\right) - 1 = 0$ | **A1 AG** | Printed result reached without error from the result of (c). Showing the printed LHS equals $\alpha^{-2}$ times the quartic, then citing (c), also scores. |
| | | **[3]** | |
| **8(e)** | $t^2 - t - 1 = 0 \Rightarrow t = \ldots$, where $t = \alpha + \frac{1}{\alpha}$ | **M1** | Solves the quadratic in $\alpha + \frac{1}{\alpha}$ by formula, completing the square or calculator. |
| | $t = \frac{1 \pm \sqrt{5}}{2}$ | **A1** | |
| | $\frac{1}{\alpha} = \cos\frac{3\pi}{5} - i\sin\frac{3\pi}{5}$<br>$\Rightarrow 2\cos\frac{3\pi}{5} = "\frac{1\pm\sqrt{5}}{2}" \Rightarrow \cos\frac{3\pi}{5} = \ldots$ | **M1** | Shows or quotes $\alpha + \frac{1}{\alpha} = 2\cos\frac{3\pi}{5}$ and equates it to their roots. |

*Question 8 continued*

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| | $\cos\frac{3\pi}{5} < 0 \Rightarrow \cos\frac{3\pi}{5} = \frac{1}{4} - \frac{1}{4}\sqrt{5}$ | **A1** | Positive value rejected as $\cos\frac{3\pi}{5} < 0$. In the form $a + b\sqrt{c}$, or $a$, $b$, $c$ stated; awrt $-0.309$ scores A0. |
| | | **[4]** | |
| | **Total for Question 8** | **12** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **9(a)** | $\alpha = \frac{(6+11i)+(-8-3i)}{2}$ | **M1** | Uses the midpoint of $z_1$ and $z_2$, with values substituted. |
| | $\alpha = \frac{-2+8i}{2} = -1+4i$ | **A1 AG** | Printed result reached from the substituted midpoint without error. |
| | | **[2]** | |
| **9(b)** | $z_1 - \alpha = 7+7i$<br>$\beta(z_1-\alpha) = \frac{(1-i)(7+7i)}{14}$ | **M1** | Finds $z_1-\alpha$ and multiplies by $\beta$. Condone a slip in $z_1-\alpha$. |
| | $= \frac{7(1-i)(1+i)}{14} = \frac{7\times 2}{14} = 1$ | **A1 AG** | Product evaluated explicitly, without error, to 1; a bare calculator result scores A0. |
| | | **[2]** | |
| **9(c)(i)** | $w = e^{\frac{k\pi}{4}i}, \quad k = 0, 1, \ldots, 7$ | **B1** | All eight roots, for any eight consecutive $k$ or listed separately. $r=1$ may be omitted. |
| | | **[1]** | |
| **9(c)(ii)** | $w = \beta(z-\alpha) \Rightarrow z = \alpha + \frac{w}{\beta}$ | **M1** | Rearranges for $z$ in terms of $w$, or otherwise rotates $z_1-\alpha$ about $\alpha$ through multiples of $\frac{\pi}{4}$. |
| | $\frac{1}{\beta} = 7(1+i)$<br>$z = -1+4i+7(1+i)w$ | **A1** | oe |
| | $w = \frac{1+i}{\sqrt{2}} \Rightarrow z = -1+4i+7(1+i)\frac{1+i}{\sqrt{2}}$<br>$w = i \Rightarrow z = -1+4i+7(1+i)i, \ldots$ | **dM1** | Substitutes at least two of the six non-real roots into their $z = \alpha + cw$, for a constant $c$. |
| | $-1+(4+7\sqrt{2})i,\ -8+11i,\ (-1-7\sqrt{2})+4i$<br>$-1+(4-7\sqrt{2})i,\ 6-3i,\ (-1+7\sqrt{2})+4i$ | **A1** | All six, exact or awrt 3 s.f., and no others; $z_1$ and $z_2$ also listed is condoned. |
| | | **[4]** | |
| **Total for Question 9** | | **9** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **10(i)** | $\lvert z\rvert = \sqrt{128^2 + (128\sqrt{3})^2}$<br>$\arg z = \pi - \arctan\sqrt{3}$ | **M1** | Attempts the modulus and an argument adjusted for the second quadrant. Condone arithmetic slips. |
| | $-128 + 128\sqrt{3}\,i = 256e^{\frac{2\pi}{3}i}$ | **A1** | |
| | $z_k = 256^{\frac{1}{8}} e^{\frac{1}{8}\left(\frac{2\pi}{3}+2k\pi\right)i}$<br>$= 2e^{\left(\frac{\pi}{12}+\frac{k\pi}{4}\right)i}$ | **M1** | Takes $r^{\frac{1}{8}}$ and $\frac{\theta+2k\pi}{8}$ from their $re^{i\theta}$, for at least two values of $k$. |
| | Two of $2e^{\frac{\pi}{12}i}, 2e^{\frac{\pi}{3}i}, \ldots$ | **A1** | Any two correct roots, arguments in any range. |
| | $2e^{\frac{\pi}{12}i}, 2e^{\frac{\pi}{3}i}, 2e^{\frac{7\pi}{12}i}, 2e^{\frac{5\pi}{6}i}$<br>$2e^{\frac{13\pi}{12}i}, 2e^{\frac{4\pi}{3}i}, 2e^{\frac{19\pi}{12}i}, 2e^{\frac{11\pi}{6}i}$ | **A1** | All eight with $r = 2$ and arguments in $0 \le \theta < 2\pi$, and no others. |
| | | **[5]** | |
| **10(ii)** |  | | |
| | *Shape* | **B1** | Eight points equally spaced on a circle centred at $O$. |
| | *Labelled point* | **B1** | $A$ to $H$ anticlockwise, $A$ at argument $\frac{\pi}{12}$ in the first quadrant. |
| | | **[2]** | |
| **10(iii)** | $w = \frac{1}{2}\left(2e^{\frac{\pi}{3}i} + 2e^{\frac{5\pi}{6}i}\right)$ | **M1** | Halves the sum of their second and fourth roots, in any form; Cartesian accepted. |

*Question 10 continued*

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| | $w = e^{\frac{7\pi}{12}i}\left(e^{-\frac{\pi}{4}i} + e^{\frac{\pi}{4}i}\right)$<br>$\lvert w\rvert = \sqrt{2}$ | **A1** | $\sqrt{2}$ or an exact equivalent. |
| | $\arg w = \frac{7\pi}{12}$ | **A1** | Or an exact equivalent. |
| | | **[3]** | |
| **10(iv)** | $w^n = (\sqrt{2})^n e^{\frac{7n\pi}{12}i}$<br>$\frac{7n\pi}{12} = m\pi,\ m \in \mathbb{Z}$ | **M1** | Requires $n$ times their $\arg w$ to be an integer multiple of $\pi$. Implied by $n = 12$. |
| | $n = 12$ | **A1** | |
| | $a = (\sqrt{2})^{12} e^{7\pi i} = -64$ | **A1** | |
| | | **[3]** | |
| | **Total for Question 10** | **13** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **11(a)(i)** | $\beta = -2 - 2i$ | **B1** | |
| | | **[1]** | |
| **11(a)(ii)** | $\alpha\beta = (-2+2i)(-2-2i) = 8$ | **B1** | Or the quadratic factor $z^2 + 4z + 8$. |
| | "8"$\gamma = -24 \Rightarrow \gamma = \ldots$ | **M1** | Sets $\alpha\beta\gamma = -24$ with their $\alpha\beta$, or divides the cubic by their quadratic factor or compares constant terms, to find $\gamma$. |
| | $\gamma = -3$ | **A1** | |
| | | **[3]** | |
| **11(a)(iii)** | $-p = -4 + "-3" \Rightarrow p = \ldots$<br>$q = 8 + "-3"(-4) \Rightarrow q = \ldots$ | **M1** | Uses $\sum\alpha = -p$ or $\sum\alpha\beta = q$ with their roots, or expands their factors $(z^2 + 4z + 8)(z - \gamma)$ and compares coefficients. |
| | $p = 7$ | **A1** | |
| | $q = 20$ | **A1** | |
| | | **[3]** | |
| **11(b)(i)** | $r = \sqrt{(-2)^2 + 2^2}$, $\theta = \pi - \arctan 1$ | **M1** | Attempts $\lvert\alpha\rvert$ and an argument in the second quadrant. Not for $\pm\frac{\pi}{4}$. |
| | $\alpha = 2\sqrt{2}\,e^{\frac{3\pi}{4}i}$ | **A1** | $r = 2\sqrt{2}$, $\sqrt{8}$ or awrt 2.83 and $\theta = \frac{3\pi}{4}$ or awrt 2.36, in the form $re^{i\theta}$; not for $r$ and $\theta$ stated alone. |
| | | **[2]** | |
| **11(b)(ii)** | $(-2+2i)^n = \left(2\sqrt{2}\,e^{\frac{3\pi}{4}i}\right)^n$<br>$= (2\sqrt{2})^n e^{\frac{3n\pi}{4}i}$ | **M1** | Applies $(re^{i\theta})^n = r^n e^{in\theta}$, or de Moivre, to their $\alpha$ in modulus-argument form from (b)(i). |
| | $= (2\sqrt{2})^n \left(\cos\frac{3n\pi}{4} + i\sin\frac{3n\pi}{4}\right)$ | **A1 AG** | Correct $\alpha^n$ from $2\sqrt{2}\,e^{\frac{3\pi}{4}i}$ leading without error to the printed result; combined steps accepted. |
| | | **[2]** | |
| **11(b)(iii)** | $\beta^n = \left(2\sqrt{2}\,e^{-\frac{3\pi}{4}i}\right)^n$<br>$= (2\sqrt{2})^n \left(\cos\frac{3n\pi}{4} - i\sin\frac{3n\pi}{4}\right)$ | **M1** | Raises $\beta$ in exponential or modulus-argument form, argument $-\frac{3\pi}{4}$ oe, to the power $n$, or takes $\beta^n$ as the conjugate of $\alpha^n$. |
| | $\alpha^n + \beta^n = 2(2\sqrt{2})^n \cos\frac{3n\pi}{4}$ | **A1** | |

*Question 11 continued*

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| | $\gamma^n = (-3)^n$<br>$\alpha^n + \beta^n + \gamma^n$<br>$= 2(2\sqrt{2})^n \cos\frac{3n\pi}{4} + (-3)^n$ | **A1 AG** | Adds $(-3)^n$ from $\gamma = -3$ to a correctly obtained $\alpha^n + \beta^n$, reaching the printed result without error. |
| | | **[3]** | |
| | **Total for Question 11** | **14** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **12(a)** | $-729 = 3^6 e^{(\pi+2k\pi)i}$<br>$\Rightarrow z = 3e^{\left(\frac{\pi}{6}+\frac{k\pi}{3}\right)i}$ | **B1** | Modulus 3 and arguments $\frac{\pi}{6}+\frac{k\pi}{3}$, soi by at least two correct roots. |
| | $A: 3e^{\frac{\pi}{6}i}$, $B: 3e^{\frac{\pi}{2}i}$, $C: 3e^{\frac{5\pi}{6}i}$<br>$D: 3e^{\frac{7\pi}{6}i}$, $E: 3e^{\frac{3\pi}{2}i}$, $F: 3e^{\frac{11\pi}{6}i}$ | **B1** | All six in exponential form; arguments may differ by multiples of $2\pi$, such as $-\frac{\pi}{6}$ for $F$. |
| | | **[2]** | |
| **12(b)** | $z_G = \frac{z_A+z_B}{3}$, $w = \frac{z_G}{z_A} = \frac{z_A+z_B}{3z_A}$ | **M1** | Forms a centroid as the mean of its three vertices, with $O$ as 0, and divides it by the corresponding root. |
| | $z_B = z_A e^{\frac{\pi}{3}i} \Rightarrow w = \frac{1+e^{\frac{\pi}{3}i}}{3}$ | **A1** | Any correct expression for $w$, oe such as $\frac{1}{2}+\frac{\sqrt{3}}{6}i$. |
| | $1+e^{\frac{\pi}{3}i} = \frac{3}{2}+\frac{\sqrt{3}}{2}i$<br>$\Rightarrow r = \ldots\ [= \sqrt{3}]$, $\theta = \ldots\ [= \frac{\pi}{6}]$ | **M1** | Finds the modulus and argument of their $w$ or its numerator, by any method such as Cartesian form or factorising out $e^{\frac{\pi}{6}i}$. Condone slips. |
| | $w = \frac{1}{\sqrt{3}}e^{\frac{\pi}{6}i}$ | **A1** | Exponential form required; oe such as $\frac{\sqrt{3}}{3}e^{\frac{\pi}{6}i}$, or awrt 0.577 and 0.524. |
| | | **[4]** | |
| **12(c)** | $p = "w"^6 \times (-729)$<br>or $p = z_G^6 = \left(\sqrt{3}\,e^{\frac{\pi}{3}i}\right)^6$ | **M1** | Multiplies the sixth power of their $w$ by $-729$, or finds the sixth power of one of their points $G$ to $L$. |
| | $p = 27$ | **A1** | |
| | | **[2]** | |
| | **Total for Question 12** | **8** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **13(i)** | $w = e^{\frac{(2k+1)\pi}{11}i}$<br>$k = 0, 1, \ldots, 10$ | **B1 B1** | **B1** Any correct general form, such as modulus-argument; $k$ need not be restricted.<br>**B1** Exactly eleven distinct roots, by a range of $k$ such as $-5$ to 5, or listed. |
| | | **[2]** | |
| **13(ii)** | $\left(\frac{z+1}{z-1}\right)^{11} = -1$<br>$\Rightarrow \frac{z+1}{z-1} = w = e^{i\theta}$, $\theta = \frac{(2k+1)\pi}{11}$ | **M1** | Divides by $(z-1)^{11}$ and sets $\frac{z+1}{z-1}$ equal to their roots from (i). Condone $z \neq 1$ not stated. |
| | $w(z-1) = z+1 \Rightarrow z = \frac{w+1}{w-1}$ | **M1** | Rearranges to make $z$ the subject. Condone sign slips. |
| | $z = \frac{e^{i\theta/2}+e^{-i\theta/2}}{e^{i\theta/2}-e^{-i\theta/2}}$<br>$= \frac{2\cos(\theta/2)}{2i\sin(\theta/2)}$ | **dM1** | Multiplies by $e^{-i\theta/2}$, or uses half-angle identities on $1 \pm \cos\theta$ and $\sin\theta$, reaching a ratio of trigonometric functions of $\frac{\theta}{2}$. |
| | $z = -i\cot\left(\frac{(2k+1)\pi}{22}\right)$ | **A1** | cso, with $\frac{1}{i} = -i$; any equivalent set of eleven values. |
| | $z$ real $\Rightarrow w$ real $\Rightarrow w = -1 \Rightarrow z = 0$<br>One real root ($m = 11$), so ten non-real | **E1** | Identifies $z = 0$ as the only real root, by $w$ real or by $\cot\left(\frac{m\pi}{22}\right) \neq 0$ for $m \neq 11$, and concludes ten non-real roots. |
| | $-i\cot\left(\pi - \frac{r\pi}{22}\right) = i\cot\left(\frac{r\pi}{22}\right)$<br>$= -i\cot\left(-\frac{r\pi}{22}\right)$<br>$m = \pm1, \pm3, \pm5, \pm7, \pm9$ | **A1 AG** | Maps $m = 13, \ldots, 21$ to $-9, \ldots, -1$ by $\cot(\pi - x) = -\cot x$ with cot odd, or by period $\pi$, and states the printed set. |
| | | **[6]** | |
| **13(iii)** | $(-i \times i)\cot^2\left(\frac{m\pi}{22}\right) = -i^2\cot^2\left(\frac{m\pi}{22}\right)$<br>$= \cot^2\left(\frac{m\pi}{22}\right)$ | **B1** | Needs $i^2 = -1$ used explicitly. |
| | | **[1]** | |
| **13(iv)** | $\cot^2\left(\frac{\pi}{22}\right)\cot^2\left(\frac{3\pi}{22}\right)\cdots\cot^2\left(\frac{9\pi}{22}\right) = 11$ | **M1** | Pairs the roots for $m$ and $-m$ and uses (iii) to equate the product of the five $\cot^2$ values to 11. |
| | $\frac{1}{11}$ | **A1** | $\frac{1}{11}$ or awrt 0.0909. |
| | | **[2]** | |
| | **Total for Question 13** | **11** | |